

LLM Provider Research

Alternatives to Groq / Llama-3.3 for the Theaurax WhatsApp Sales Bot

Prepared 15 July 2026 · Pricing in INR at USD 1 = ₹96

Executive Summary

This report evaluates candidate replacements for the bot's current Groq / Llama-3.3 primary provider, with particular attention to two requirements the current stack does not satisfy well: a viable free or low-cost tier that survives real concurrent traffic, and high-quality Tanglish (romanized Tamil-English code-mixed) generation.

Recommendation

Route Tanglish sessions to Sarvam-30B and keep Groq / Llama-3.3 as the English primary. Estimated cost is approximately ₹360/month at 100 conversations/day — less than the margin on a single jersey sale. Do NOT use NVIDIA NIM's free hosted endpoint for this bot: it is licensed for prototyping only, and serving real customers on it is a terms violation.

Three findings drive this conclusion. First, Cerebras' free tier caps at 1M tokens/day, which this bot exhausts in roughly 28 conversations — the same class of ceiling that ended the Qwen3 experiment in July 2026. Second, NVIDIA's free endpoint is explicitly scoped to prototyping; production requires an NVIDIA AI Enterprise licence starting at USD 4,500 per GPU per year. Third, Sarvam AI — an Indian provider absent from the original shortlist — was purpose-trained on romanized and code-mixed Indian-language text, supports OpenAI-compatible tool calling, bills in rupees, and is materially cheaper than every alternative assessed.

1. Measured Baseline of the Existing System

All cost projections in this report are anchored to measurements taken from the running codebase on 15 July 2026, not to vendor-supplied averages. The bot's per-request cost is bounded by three hard limits already present in `src/services/ai.js`.

Constraint	Measured value	Source
Input per call (ceiling)	20,000 chars (~5,700 tok)	<code>trimMessagesToTokenBudget()</code> , <code>ai.js:346</code>
Output per call (ceiling)	800 tokens	<code>max_tokens: 800</code> , <code>ai.js:431</code>
Loops per user message	5	<code>while (keepLooping && loops < 5),</code> <code>... call ...</code>
System prompt	6,549 chars (~1,871 tok)	<code>generateSystemPrompt()</code> , measured
Tool definitions	2,382 chars (~681 tok)	<code>getTools()</code> , sent every call
Product cache	136 products, ~830 chars ea.	<code>products_cache.json</code> , measured

The practical consequence is a fixed floor of roughly 2,550 tokens on every single call — the system prompt plus tool schema — before any conversation content is added. A first-turn call runs about 3,000 tokens; a late-turn call carrying history and a search result approaches the ~6,400-token ceiling.

Revised per-conversation estimate

An earlier working estimate of 25,000 input tokens per conversation was too low. It incorrectly treated the 5-loop cap as per-conversation, when in fact the loop is per user message — `answerQuery()` runs its own loop on each inbound message. After the deterministic fast paths absorb size/quantity parsing and order confirmation, a realistic purchase conversation reaches the LLM on roughly 4–6 messages, averaging about 1.7 loops each (one to invoke the tool, one to narrate the result).

Metric	Value	Basis
LLM calls per conversation	~8	4-6 msgs x ~1.7 loops
Average input per call	~4,500 tokens	floor 2,550 + history/results
Input per conversation	~36,000-40,000 tok	revised upward from 25,000
Output per conversation	~2,000 tokens	~250 real tok/reply x 8

This report uses 40,000 input / 2,000 output tokens per conversation. Note the ranking of providers is unchanged by this 1.6x revision — the recommendation is robust even if the true figure is double the estimate.

2. Free Tier Analysis

Every free tier assessed fails for this workload, for a different reason in each case. This is the central negative finding of the research.

Provider	Free limits	Verdict	Reason
Cerebras	5 req/min, 30K tok/min, 1M tok/day (per	No	~28 conversations/day
NVIDIA NIM	~40 req/min, no daily token cap	No	Prototyping licence only
Mistral	~2 req/min, 1B tok/month	No	2 RPM unusable
OpenRouter	20 req/min, 50 req/day (1,000 if \$10 spent)	No	50/day is negligible
Sarvam	₹1,000 signup credit, 60 req/min	Part	Credit, not a free tier

Why Cerebras fails

The Cerebras console screenshot shows 1,000,000 tokens/day as a hard ceiling. At the measured ~36,000–40,000 tokens per conversation, that is approximately 28 conversations per day before the quota is gone. The 30,000 tokens/minute limit is equally binding: it permits only 5–6 calls per minute shared across all concurrent customers.

One legitimate mitigation exists. The quota shown in the console is applied per model, not per account. Rotating across gpt-oss-120b, gemma-4-31b and zai-glm-4.7 yields 3M tokens/day — roughly 85 conversations. The existing rotateEntries() key-rotation logic is already the right shape for this. That makes Cerebras defensible as a fallback tier, but not as a primary.

Pattern worth noting

This is structurally the same failure that ended the Qwen3 experiment: adequate quality, but a free-tier throughput ceiling that cannot absorb concurrent traffic. Cerebras' 30K tok/min is better than Qwen's 8K, yet the 1M/day cap binds first. Free-tier ceilings on this bot are a token-volume problem, not a request-rate problem.

3. NVIDIA NIM — Detailed Assessment

NVIDIA warrants separate treatment because its free tier is superficially the most attractive option assessed — roughly 40 requests/minute with no published daily token cap, across 100+ models on an OpenAI-compatible endpoint — and because the disqualifying constraint is contractual rather than technical.

Three distinct delivery modes

Mode	Cost	Permitted use
Hosted API (build.nvidia.com)	Free	Prototyping and evaluation only
Downloadable NIM container	Free from NVIDIA; you pay GPU	Research/dev, up to 16 GPUs
NVIDIA AI Enterprise	From \$4,500/GPU/yr (~₹4.3L)	Production, real end users

Disqualifying finding

NVIDIA's own developer page describes the free hosted endpoints as being for "unlimited prototyping." Serving real end users or business transactions is classed as production and requires an NVIDIA AI Enterprise licence. A WhatsApp bot taking live jersey orders is unambiguously production, so the free endpoint is not a lawful option for this deployment regardless of whether the rate limits would technically suffice.

The practical cost of compliance makes this decisive. NVIDIA AI Enterprise begins at approximately USD 4,500 per GPU per year — about ₹4,32,000 — or roughly USD 1 per GPU-hour in the cloud (~₹96/hour, i.e. ~₹69,000/month if run continuously). Against a projected ₹360/month on Sarvam, the enterprise path is over 1,000x the cost for a bot at this volume.

Models observed in the NVIDIA catalogue

The following were visible on build.nvidia.com and are noted for completeness. All are marked "Free Endpoint" and "Downloadable" — the free-endpoint badge refers to the prototyping tier described above.

- glm-5.2 (Z.ai) — flagship model for agentic workflows, coding and long-horizon reasoning.
- nemotron-3-ultra-550b-a55b (NVIDIA) — hybrid Mamba-Transformer MoE, 1M context, agentic reasoning.
- kimi-k2.6 (Moonshot AI) — 1T multimodal MoE for long-horizon coding, agentic tool use, image/video.
- deepseek-v4-pro (DeepSeek AI) — MoE architecture scaling to 1M-token context, coding-oriented.

None of these is Indic-specialised, so none addresses the Tanglish requirement even setting the licensing question aside.

Third-party hosting of deepseek-v4-pro

NVIDIA's model page lists external providers who serve the same weights commercially. These are legitimate production options and are converted to INR below. OpenRouter is by a wide margin the cheapest route to this model.

Provider	Input \$/M	Output \$/M	Input ₹/M	Output ₹/M	Dedicated GPU
OpenRouter	\$0.44	\$0.87	₹42.24	₹83.52	—
Lightning AI	\$0.89	\$3.00	₹85.44	₹288.00	—
GMI Cloud	\$1.13	\$2.26	₹108.48	₹216.96	from \$0.53/min
Deep Infra	\$1.30	\$2.60	₹124.80	₹249.60	from \$0.01/min
Bitdeer AI	\$1.39	\$2.78	₹133.44	₹266.88	—
Together AI	\$1.74	\$3.48	₹167.04	₹334.08	from \$1.20/min

4. Pricing in Indian Rupees

Converted at USD 1 = ₹96 (mid-July 2026). Monthly projections assume 100 conversations/day (3,000/month) at the measured 40,000 input / 2,000 output tokens per conversation — that is 120M input and 6M output tokens per month.

Model	Input ₹/M	Output ₹/M	Est. ₹/month	Notes
Sarvam-30B	₹2.50	₹10.00	₹360	Tanglish-native
Ministral 3B	₹3.84	₹3.84	₹484	Very small model
Sarvam-105B	₹4.00	₹16.00	₹576	Tanglish-native,
Tencent Hy3 preview	₹6.05	₹20.16	₹847	262K ctx, 2 providers
DeepSeek V4 Flash	₹8.64	₹17.28	₹1,141	1.05M ctx, 17
Gemini 2.5 Flash Lite	₹9.60	₹38.40	₹1,382	1.05M ctx
Mistral Small 3.1	₹19.20	₹57.60	₹2,650	—
Cerebras gpt-oss-120b	₹33.60	₹72.00	₹4,464	Paid Developer tier
DeepSeek V4 Pro (OR)	₹42.24	₹83.52	₹5,570	Cheapest V4 Pro
Cerebras zai-glm-4.7	₹216.00	₹264.00	₹27,504	Not cost-competitive

Two observations. First, Cerebras' paid tier costs roughly 12x Sarvam-30B for this workload. Cerebras sells throughput — on the order of 3,000 tokens/second — which a WhatsApp bot does not need; a customer cannot perceive the difference between a 200ms and a 2-second reply on a messaging app. Second, Sarvam is cheaper than every USD-denominated competitor largely because it is priced for the Indian market in rupees rather than converted from dollars.

Prompt caching

The system prompt and tool schema together total 8,931 chars — a byte-identical prefix on every request, and roughly 57% of a typical call's input. Sarvam bills cached input at ₹1.50/M against ₹2.50/M fresh, which would reduce the Sarvam-30B projection from about ₹360 to roughly ₹290/month. The saving is modest in absolute rupees but effectively free, since the prefix is already constant and requires no code change beyond enabling caching.

5. Tanglish Capability — Sarvam AI

Sarvam AI is an Indian provider building sovereign Indic-language models. It did not appear in the original candidate set but is the strongest match for the requirement that motivated both the Gemini routing decision and the reverted Qwen3 experiment.

Why it fits

- Sarvam-30B and Sarvam-105B were explicitly trained on native-script, romanized AND code-mixed inputs across the ten most-spoken Indian languages, Tamil included. Romanized colloquial text is a first-class evaluation category, not an emergent side-effect of multilingual pretraining.
- Sarvam's own positioning cites handling code-mixed Hinglish/Tanglish "that global APIs fumble" — precisely the failure mode observed with Llama-3.3.
- Tool calling is supported via an OpenAI-compatible /chat/completions endpoint with tools, tool_choice, and finish_reason: "tool_calls". This was the specific capability that constrained the Qwen3 attempt, and it is the hard requirement for the existing agentic loop.
- Rate limit is 60 requests/minute on the free Starter plan — twelve times Cerebras' 5 RPM — with no daily token ceiling.
- Billing is in INR with credits that do not expire. Both models are open-weight, leaving a self-hosting path available later.

Migration note

Sarvam-M (24B) is deprecated and no longer available through the Chat Completions API. Any blog post or tutorial referencing sarvam-m is stale. Use Sarvam-30B or Sarvam-105B.

Cost of entry

At ₹2.50/M input, the ₹1,000 signup credit alone covers roughly 12,000 conversations before any payment is required. One documented discrepancy should be noted: Sarvam's docs pricing page states ₹100 in free credits while the plans page states ₹1,000 plus a ₹300 Starter bonus. Either figure provides meaningful runway, but the higher number should not be relied upon until confirmed in the account console.

6. System Findings

Three issues in the existing codebase were identified during this research. None blocks the provider decision, but each has independent value.

Finding 1 — System prompt has grown 45% beyond its documented size

CLAUDE.md records that the prompt was cut "from ~7000 to ~4500 JSON chars" when the FAQ block was removed. It measures 6,549 chars today. The FAQ removal did hold — lines 129-134 are now only a brief pointer to the code-level matcher. The growth came afterward, from accumulated guardrails: the WORKED EXAMPLES block, the TOOL FORMAT section, and rules 10 through 12.

This is not bloat to be cut blindly. Each addition traces to a real defect — rule 10 exists because the model was dropping the payment URL and customers could not pay; rule 12 exists because it was confirming orders on non-confirmations. The documentation is simply stale relative to the code.

Finding 2 — Worked examples are sent in both languages regardless of session language

The WORKED EXAMPLES block is 2,681 chars — 41% of the entire system prompt, roughly 766 tokens on every call. It contains both English and Tanglish variants, yet session.language is already deterministically decided and locked before the prompt is generated. Emitting only the relevant language's examples would cut roughly 1,000–1,600 chars (~290–460 tokens) from every request with no behavioural change, since the model only ever needs the examples for the language it has been instructed to reply in.

Finding 3 — Rupee symbol likely renders blank on customer invoices

src/services/invoice.js prints the ₹ character at lines 99, 100, 118, 125, 132 and 147 without registering a TrueType font. PDFKit's built-in Helvetica uses WinAnsi encoding, which has no glyph for U+20B9; measurement confirms widthOfString("₹") returns 0 under Helvetica, against 6.67 under Arial. Every rupee symbol on the branded proforma invoices sent to customers is therefore likely rendering as blank space. The fix is one registerFont call — the same technique used to produce this report.

7. Recommendation

Traffic	Provider	Rationale
Tanglish	Sarvam-30B	Purpose-trained on romanized code-mixed Tamil
English	Groq / Llama-3.3	Already proven at production volume; keep
Fallback	OpenAI GPT-4o-mini	Existing chain, unchanged
Optional 2nd	Cerebras (3-model rotation)	3M tok/day free if a spare tier is wanted
Deprecate	Gemini	limit: 0 unresolved; Sarvam supersedes its role

This closes the open Gemini decision recorded in CLAUDE.md without requiring Google Cloud billing to be enabled. Sarvam is both cheaper than Gemini would have been and better at the specific task Gemini was being retained for. The projected cost is approximately ₹360/month at 100 conversations/day, or roughly ₹290 with prompt caching enabled.

Suggested sequence

- Add Sarvam as a provider in ai.js, gated behind SARVAM_API_KEY, routed for session.language === "tanglish". The endpoint is OpenAI-compatible, so the existing agentic loop should need only a base-URL and key swap.

- Add per-call token logging at the same time. This replaces the 40,000-token estimate in this report with measured production figures and should be done before committing to any volume-based plan.
- Split the worked examples by session language (Finding 2) — an immediate ~290-460 token saving per call on every provider, including the current Groq path.
- Fix the invoice rupee glyph (Finding 3) — independent of the provider decision, but customer-facing.
- Update CLAUDE.md to reflect the true 6,549-char prompt size (Finding 1).

8. Caveats

- The 40,000 tokens/conversation figure is derived from measured prompt sizes and code-enforced ceilings, not from production telemetry. It should be validated with real token logging before any volume commitment. The provider ranking is unchanged even at 2x this figure.
- Sarvam's free-credit amount is documented inconsistently (₹100 on the docs pricing page, ₹1,000 plus ₹300 bonus on the plans page). Confirm in the account console.
- NVIDIA's prototyping-versus-production boundary is stated on its developer pages and in secondary sources rather than in a single quotable clause of the licence. If NVIDIA is seriously considered, obtain written confirmation from NVIDIA before serving customer traffic.
- Cerebras per-model quota separation is inferred from the console screenshot, which lists quota per model row. It has not been empirically verified that the three models draw from independent daily buckets.
- Sarvam's Tanglish quality claims are drawn from vendor documentation and the model announcement, not from independent benchmarking on this bot's traffic. A shadow test against real conversations is advisable before full cutover.

Sources

- Cerebras rate limits — inference-docs.cerebras.ai/support/rate-limits
- Cerebras pricing — cerebras.ai/pricing
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- Sarvam models — docs.sarvam.ai/api/getting-started/models.md
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- Codebase measurements: E:\TheAurax\src\services\ai.js, invoice.js, products_cache.json — 15 July 2026